Data Visualization Portfolio

From Prompt to Plot — Data Visualization with AI Copilots

Alexander Weber

2026-04-28

Task 1 — Critique of a Manipulative Visualization

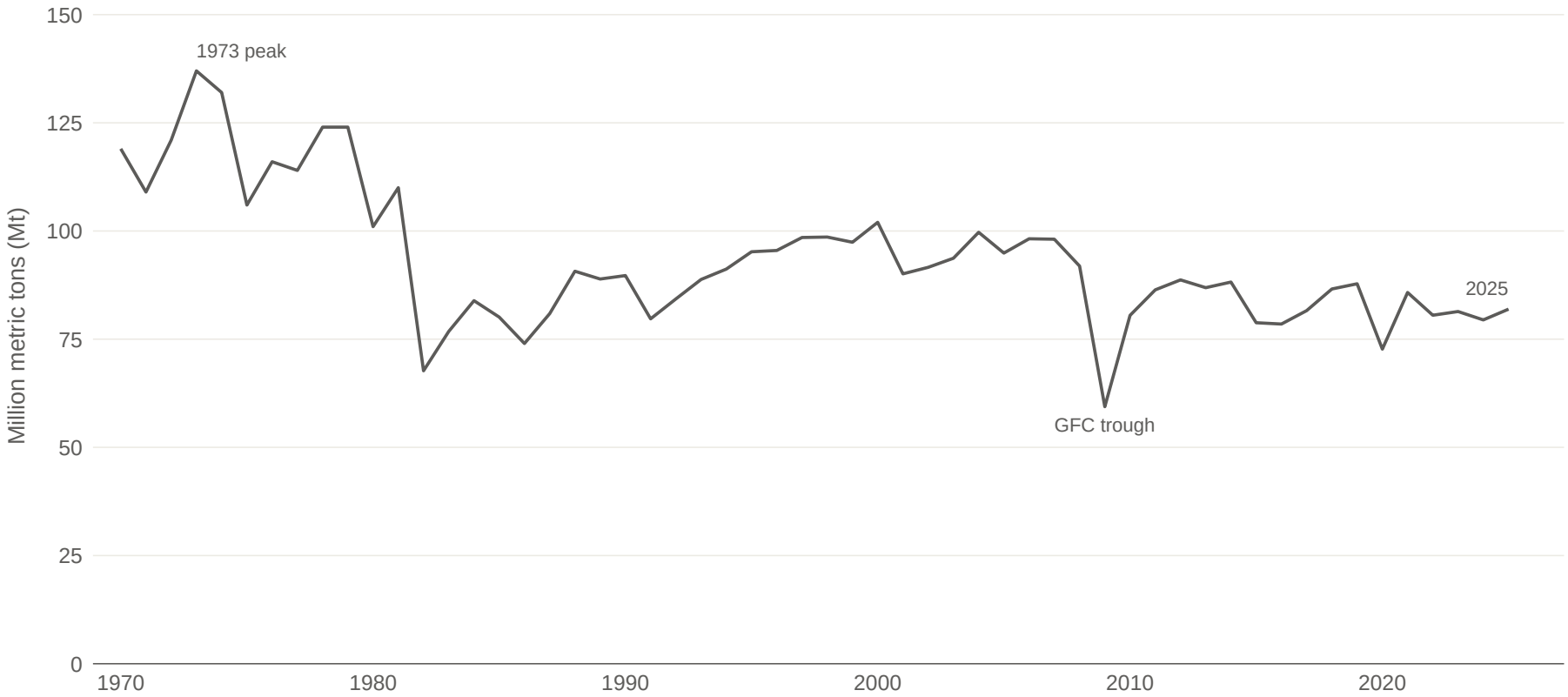
Source: https://x.com/WhiteHouse/status/2017370992436249025

The most obvious issue is the truncated y-axis, which exaggerates the perceived change in steel production and creates an immense lie factor. On top of overstating the change, the chart also uses color encoding to further mislead the viewer. They make the fill of the prior year red and that of 2025 green, which in western societies conveys positive sentiment. They also add a green font to the "INCREASES" to really convey a positive development. They additionally cherry-pick a two-year comparison, which is not representative of the historical range of US steel production. Two data points are not enough to tell a story. A minor issue I don't like either is the naming of the x-axis, where in one instance the year is at the start and in another at the end.

Task 2 — Improved Version

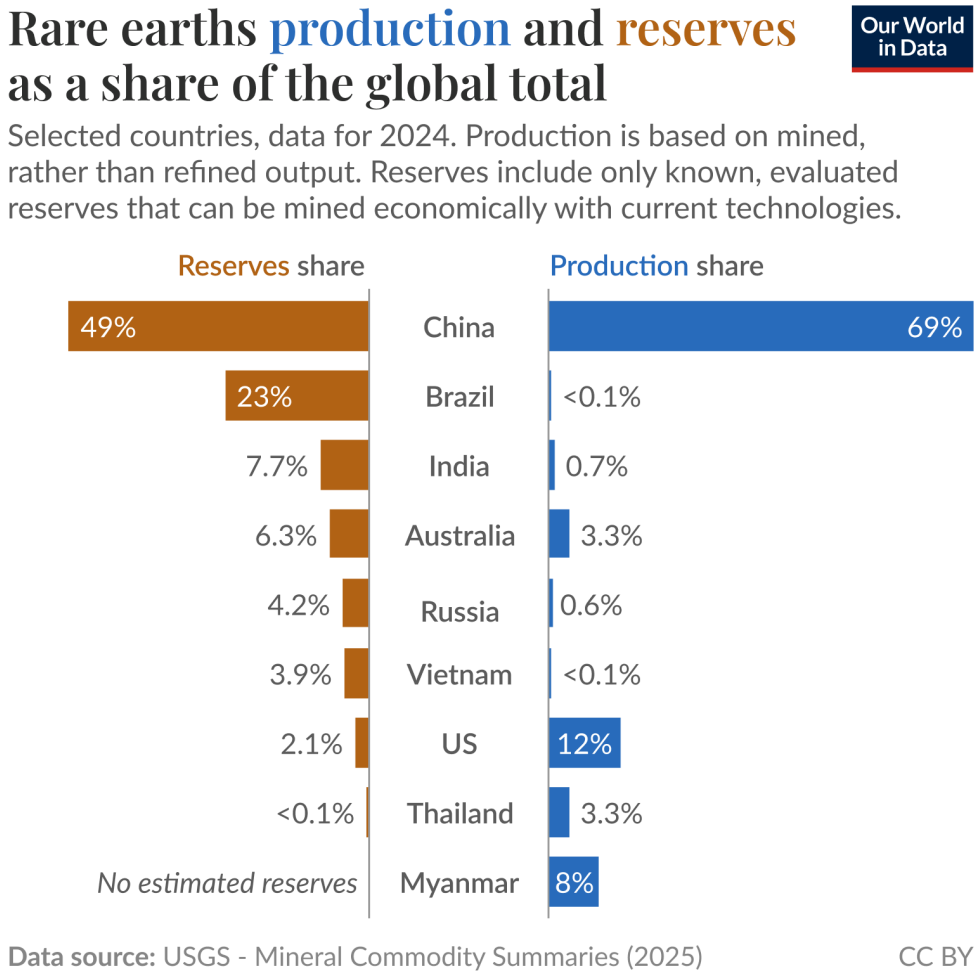
US raw steel production is 41% below its 1973 peak — and the 2024–2025 increase is noise.

Annual US raw steel production, 1970–2025. Output peaked in 1973 and roughly halved by the mid-1980s; it has plateaued near 80 Mt since 2010. The 2024–2025 increase falls well inside the year-to-year noise of every decade since.



Sources: USGS Data Series 140, "Iron and Steel Statistics" (1900–2021); World Steel Association, P1 crude steel total, USA (2021–2025).
https://www.usgs.gov/centers/national-minerals-information-center/historical-statistics-mineral-and-material-commodities
https://worldsteel.org/data/annual-production-steel-data/

Task 3 — Critique of a Particularly Good Visualization



Source: https://ourworldindata.org/data-insights/brazil-india-vietnam-and-russia-hold-large-reserves-of-rare-earth-but-mine-very-little-of-them

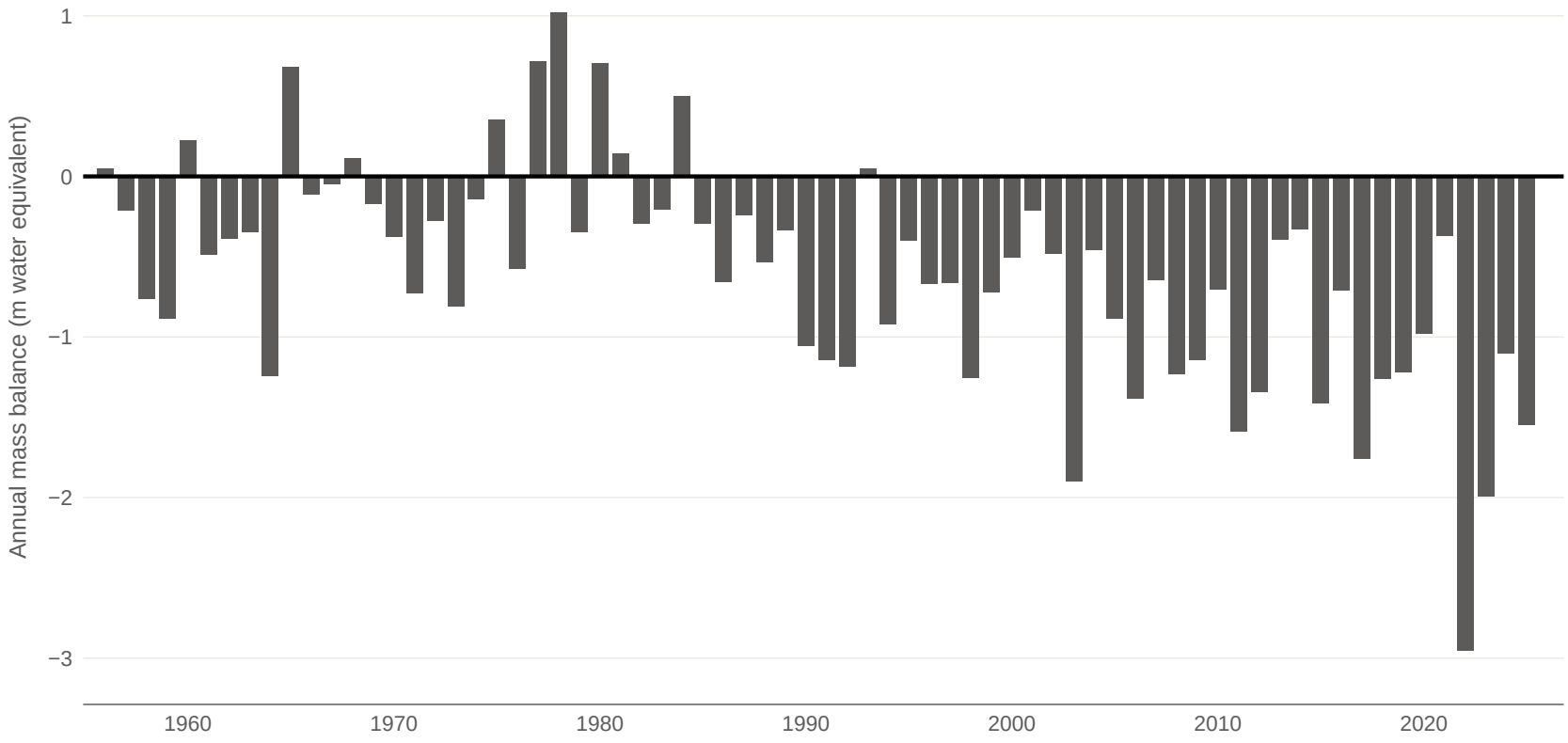
This is a good visualization because it tells a clear story. This story is supported by the fact that the maker chose to sort the reserves by descending order, making the disparity between reserves and production immediately apparent. The chart uses color, but in a good way that makes the distinction between the two variables easy to see. The axis is not truncated and the bars lengths faithfully represent the data.

There are some minor issues worth mentioning though. The visualization uses "selected countries" for the visualization, which leaves me to question what criteria determined which countries were included. The graph also shows percentages <0.1% which then are not truthful to the size of the bars. Finally, they include a measure of a country that does not have recorded reserves. This makes it questionable whether this is a valid data point.

Task 4 — Climate-change Visualization

Swiss glaciers have lost mass every year for 32 straight years.

Annual mass balance across Switzerland's monitored glaciers, area-weighted, 1956–2025. The annual loss rate has roughly tenfolded since the 1960s; recent years (2022–2025) average -1.9 m water equivalent per year against -0.18 for 1956–1989. Three near-zero years (1956, 1967, 1993) are rendered at minimum visible height so no observation reads as a missing year.

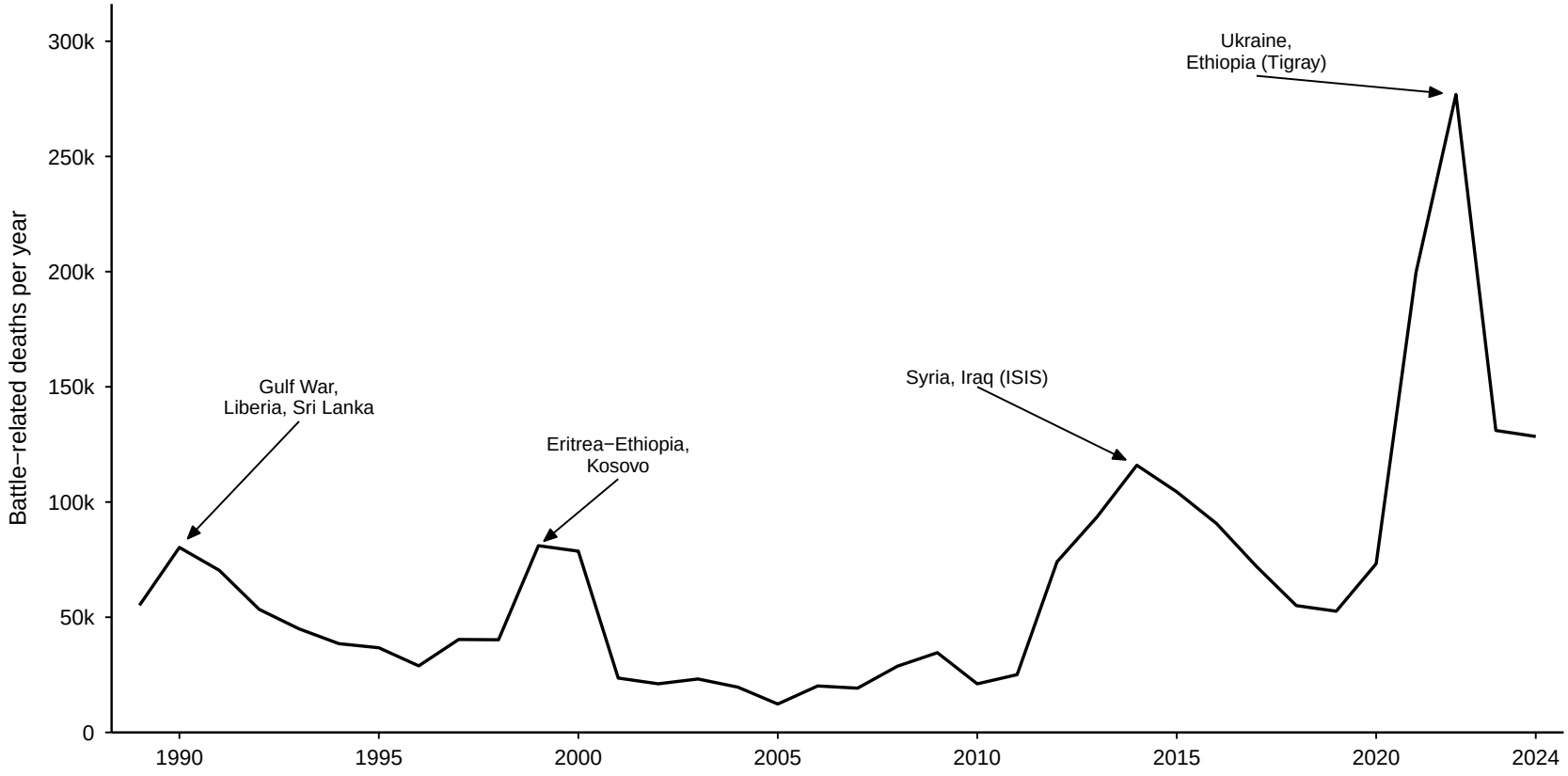


Source: GLAMOS (2025), Swiss Glacier Mass Balance, release 2025. https://doi.glamos.ch/data/massbalance/massbalance_2025_r2025.html

Task 5 — Black-and-white Visualization

The four deadliest years in 35 years of UCDP records are all 2021–2024.

Global battle-related deaths in state-based armed conflicts, 1989–2024 (UCDP best estimate). The 2022 peak (Ukraine + Tigray) is the highest annual value the dataset has recorded; 2021, 2023, and 2024 round out the top four.

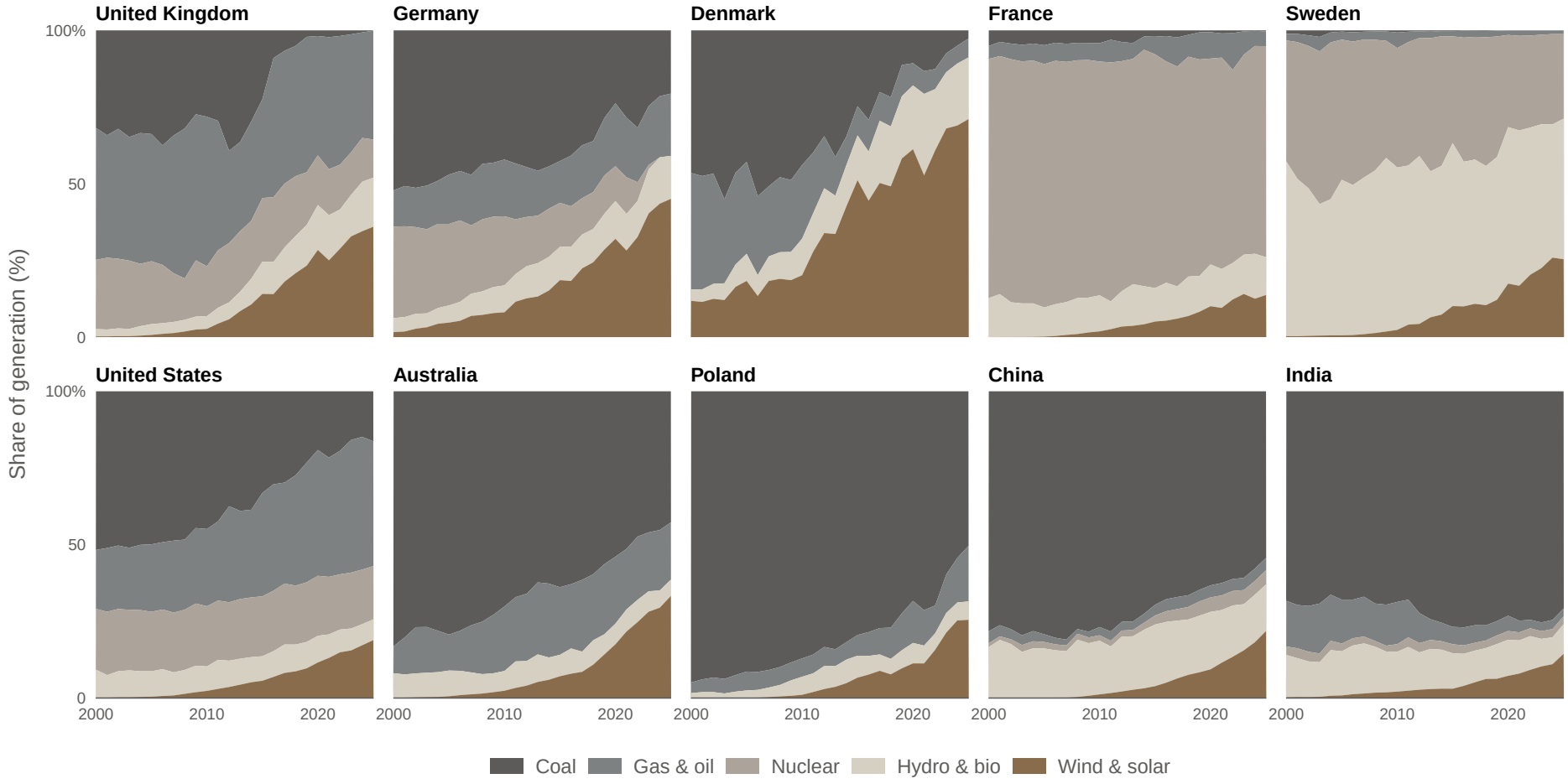


Source: UCDP (2025), Battle-Related Deaths Dataset v25.1, Uppsala Conflict Data Program. https://ucdp.uu.se/downloads/

Task 6 — Color as an Important Aesthetic

Wind and solar replaced coal in Europe — but China and India are still coal-anchored.

Share of electricity generation by fuel category, ten countries, 2000–2025. Sweden and France remain low-carbon throughout (nuclear, hydro). Germany, Denmark, and the UK rebuilt their grids on wind and solar. Coal still supplies more than half of generation in China, India, and Poland.

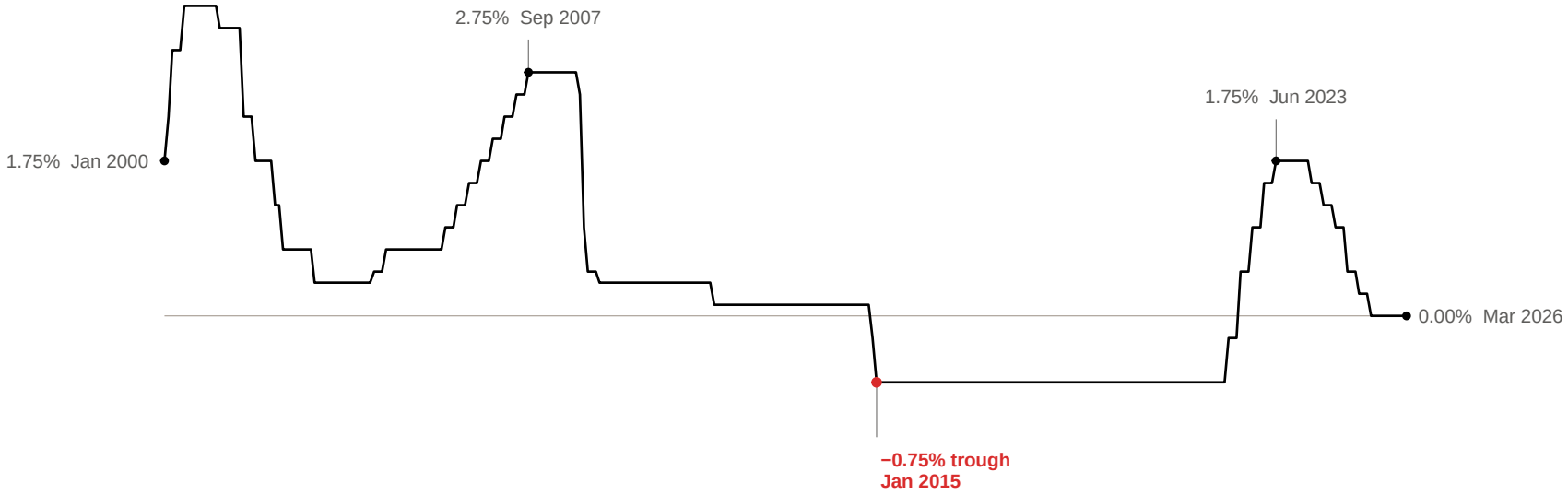


Source: Ember (2025), Yearly Electricity Data. https://ember-energy.org/data/yearly-electricity-data/

Task 7 — Maximum Data-ink Ratio (Tufte)

The SNB cut its policy rate back to zero in March 2026.

Swiss National Bank policy rate, monthly, 2000–2026. Pre-GFC peak 2.75% (Sep 2007); eight years below zero (Dec 2014 – Sep 2022, trough -0.75%); a brief hike to 1.75% (Jun 2023); and now back to zero — the lowest setting since exit.

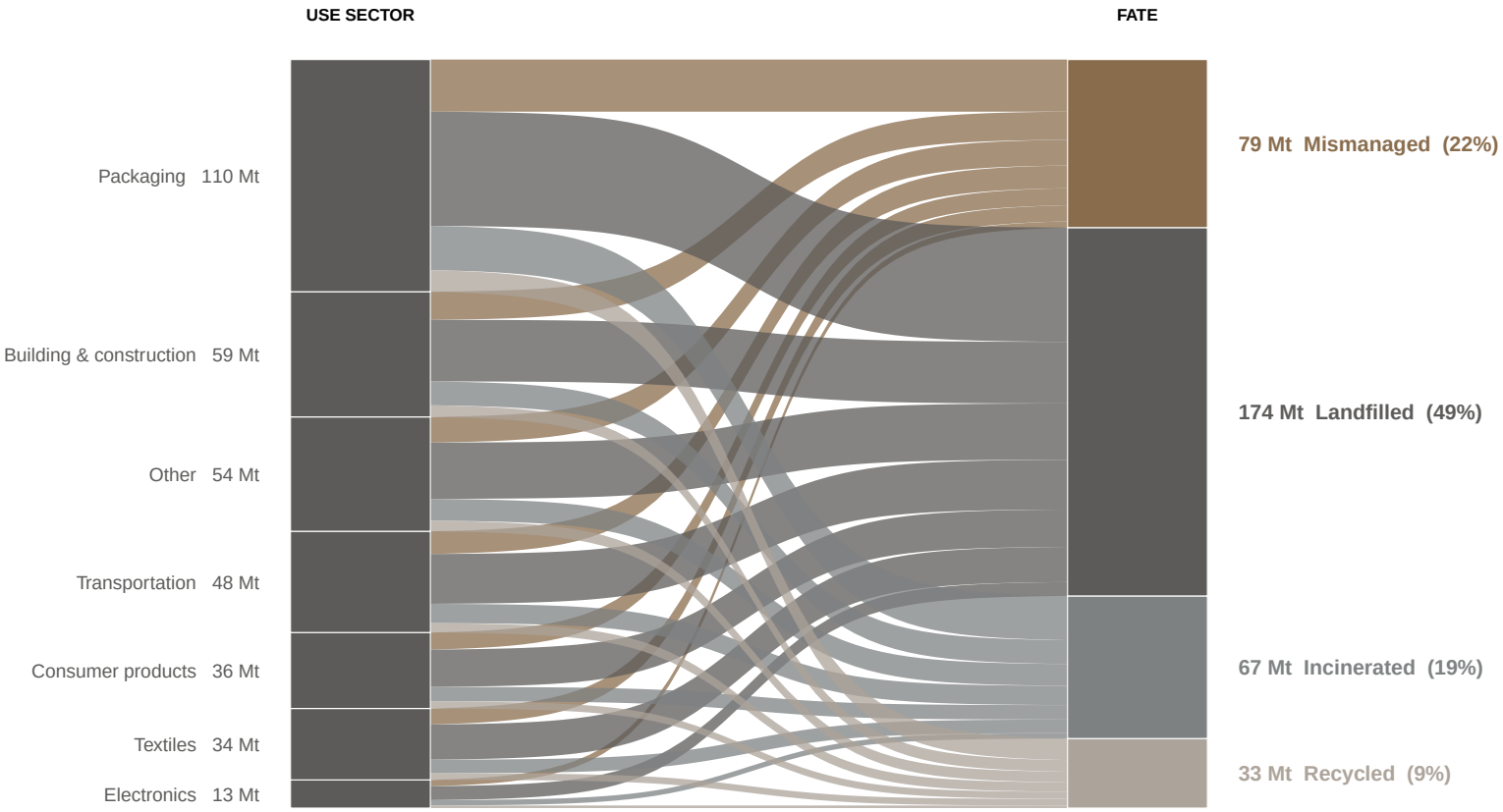


Source: Swiss National Bank, "Official rates of the SNB" (cube snboffzisa). Spliced: 3-month CHF LIBOR target band mid (pre-Jun 2019) + SNB Leitzins (Jun 2019+).
https://data.snb.ch/

Task 8 — Non-standard Chart Type

Less than 10% of the world's plastic waste is recycled.

Global plastic waste reaching end-of-life in 2019, by use sector and fate. Of 353 Mt of post-use waste, only 33 Mt — 9% — was recycled. Half went to landfill; almost a quarter (79 Mt) leaked into the environment as mismanaged waste.

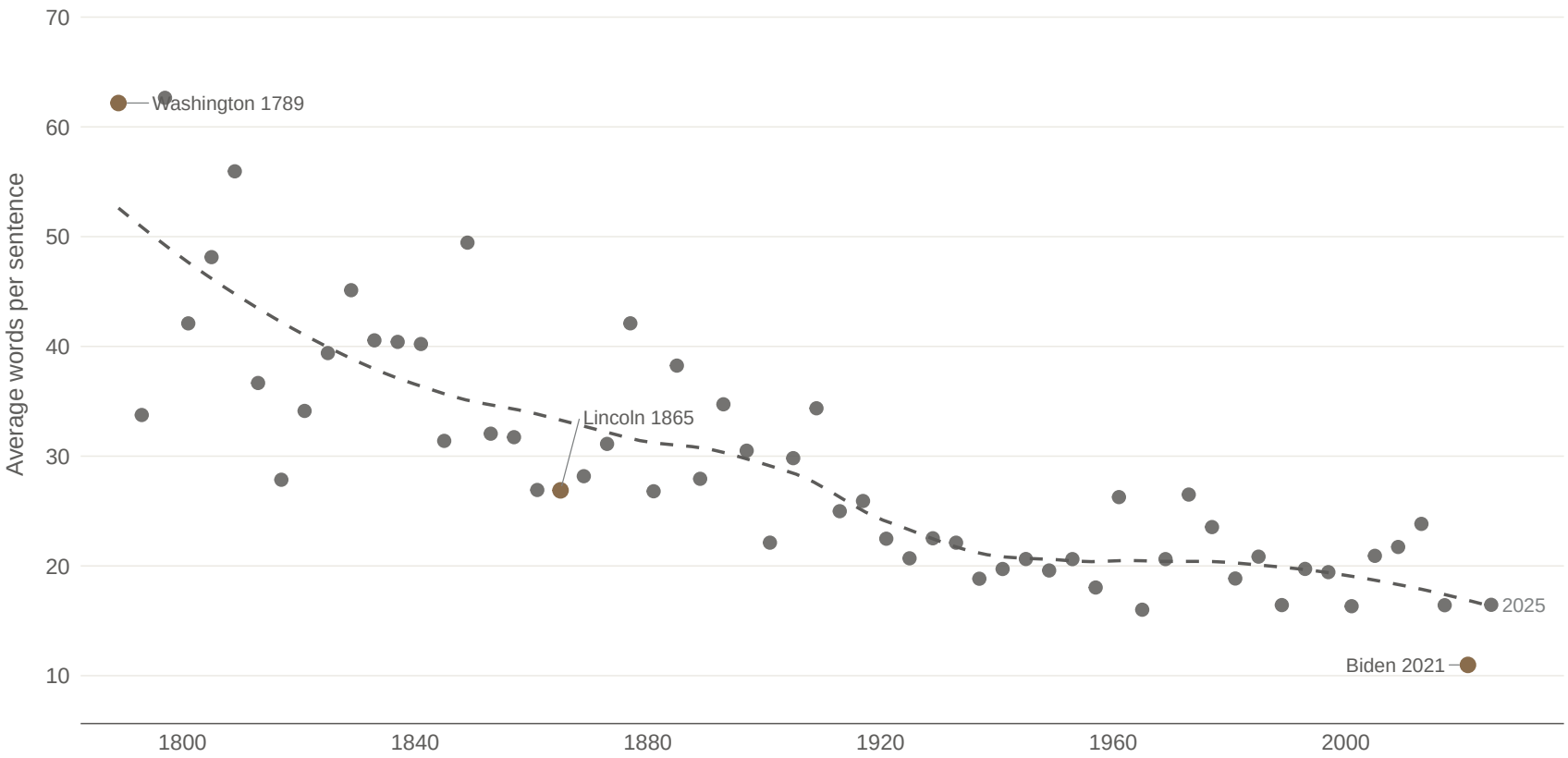


Source: Our World in Data, Plastic Pollution *(extending Geyer, Jambeck & Law 2017, Science Advances).* *https://ourworldindata.org/plastic-pollution*

Task 9 — Visualization of Textual Data

American inaugural sentences have shrunk from over 60 words to under 20 in 235 years.

Average words per sentence in each US presidential inaugural address, 1789–2025. Washington opened with 62 words per sentence (Latinate, periodic prose); Biden's 2021 address ran at 11 — the lowest in the dataset. The decline is structural, not just shorter speeches: long, clause-stacked sentences disappeared from the lectern, replaced by short, conversational lines.



Source: US presidential inaugural addresses, 1789–2025, via the `quanteda` R package corpus `data_corpus_inaugural` (compiled from Bartleby.com and the Miller Center / American Presidency Project). Sentence and word counts computed with quanteda's tokenizer. https://millercenter.org/the-presidency/presidential-speeches

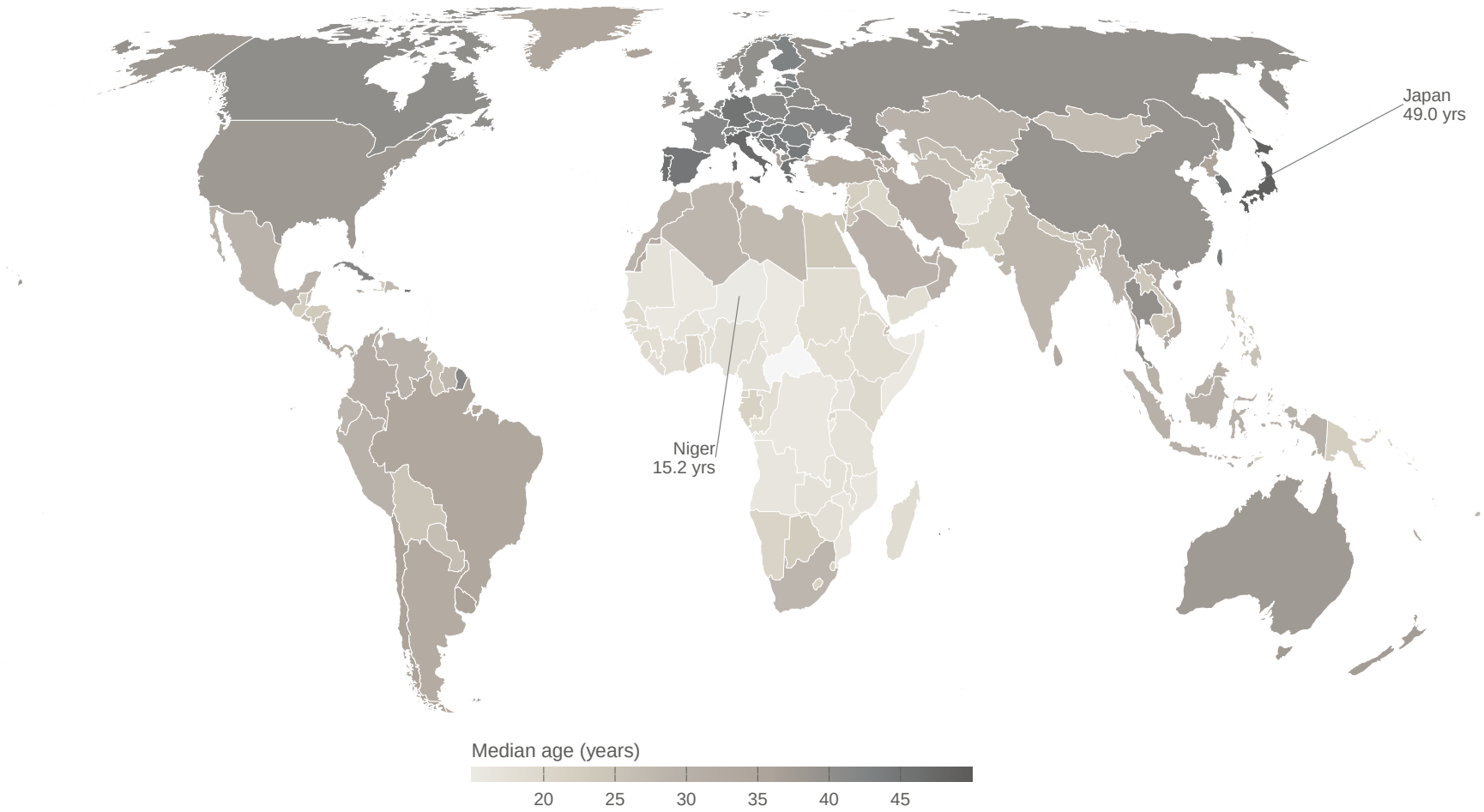
Task 10 — Hand-drawn Visualization

Hand-drawn visualization to be inserted here.

Task 11 — Data Map

The world's median age splits cleanly along the demographic transition.

Median age by country, 2023. Sub-Saharan Africa runs under 20; North America and most of Europe sit above 38. Italy, Germany, Japan, and South Korea are all above 45 — and aging fast. Equal-area Eckert IV projection so country size on the page equals country size on the ground.



Source: UN Department of Economic and Social Affairs, Population Division, World Population Prospects 2024 *(medium variant, estimates), via Our World in Data.*
https://ourworldindata.org/grapher/median-age

Task 12 — Interactive: 45 Years of Earthquakes

Where the Earth's plates actually grind: 45 years of M$\geq$5 earthquakes.

Year-range slider, magnitude threshold, region focus, and click-a-hex for the top events in that region. The Pacific Ring of Fire, Mediterranean–Himalaya belt, Mid-Atlantic Ridge, and East African Rift emerge only when 45 years of seismicity are allowed to accumulate. Hosted on GitHub Pages via Observable Framework.

Where the Earth's plates grind

45 years of M≥5 earthquakes from the USGS ANSS catalog, 1980–2025. Drag the year and magnitude controls, switch the projection focus, and click a hex on the map to inspect the largest events in that region.

Controls

Start year 1980

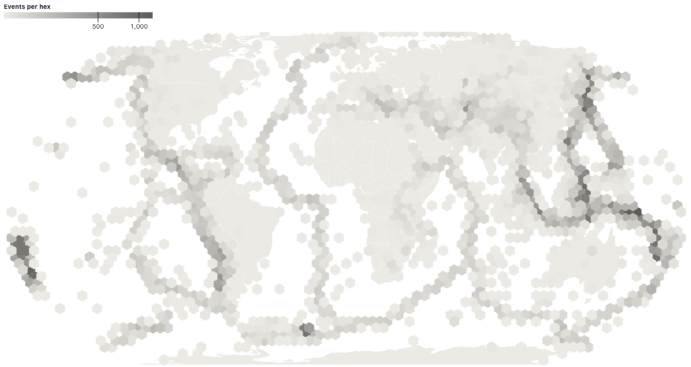
End year 2025

Minimum magnitude 5

Region World

Reset all filters

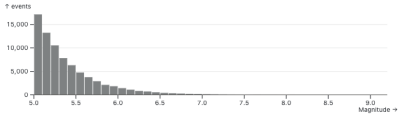
Map



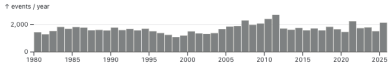
Selected hex — top events

Click a hex on the map to inspect the largest events in that region.

Magnitude distribution



Events per year



Data. USGS Earthquake Hazards Program, Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat). Queried via the FDSN event web service: earthquake.usgs.gov/fdsnws/event/1/. All events with magnitude ≥ 5.0 between 1980-01-01 and 2025-12-31 (~75,000 events). USGS preferred magnitudes are heterogeneous across the catalog (Mw, mb, ML, Ms); for visualization purposes the catalog is treated as a single magnitude axis. Pre-1980 catalog completeness in remote ocean basins is materially lower than post-1990 due to broadband seismograph network expansion — starting at 1980 avoids that caveat without losing analytical content. A small fraction (~2%) of place names from the upstream feed contain a literal "?" where a non-ASCII character was lost during USGS ingestion (e.g. "?funato" for "Ōfunato"); these are preserved unchanged here.

Basemap. Natural Earth countries, 110m resolution, decoded from world-atlas TopoJSON via topojson-client. Equal-Earth projection.

Tooling. Built with Observable Framework, Observable Plot, and D3. Cell scaffold and iteration assistance from Claude Code (Opus 4.7, with the Playwright and Context7 MCP servers). Deployed to GitHub Pages via Actions.

Built with Observable on Apr 27, 2026.

Try it live: https://alexander-the-sufficient.github.io/data_vizualisation_with_ai/task_12/

Source: USGS Earthquake Hazards Program, ANSS Comprehensive Earthquake Catalog (ComCat), via the FDSN event web service. https://earthquake.usgs.gov/fdsnws/event/1/